

ORIGINAL PAPER

AYURVEDA: A HISTORICAL PERSPECTIVE
AND PRINCIPLES OF THE TRADITIONAL
HEALTHCARE SYSTEM IN INDIA

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Ayurveda, the science of life, is a comprehensive medical system that has been the traditional system of healthcare in India for more than 5000 years. This medical system was well established around 2500 to 600 BC, when it evolved into 2 schools: the School of Physicians and the School of Surgeons, similar to allopathy. Charak Samhita, Susrut Samhita, and Ashtang Hridaya Samhita are the Senior Triad texts, and Madhav Nidan Samhita, Sarangdhar Samhita, and Bhavprakash Samhita are the Junior Triad texts. Around 600 BC, Ayurveda was branched into internal medicine; pediatrics; psychiatry; surgery; eye, ear, nose, and throat; toxicology; geriatrics; and eugenics/aphrodisiacs. The body is composed of 3 body doshas, 3 mental doshas, 7 dhatus, and malas. The harmony among the body doshas of vata (nervous system), pitta (enzymes), and kapha (mucus) and the gunas, or mental doshas (which are human attributes: satogun [godly], rajas [kingly], and tamas [evil]), constitutes health, and their disharmony constitutes disease. The management of illness requires balancing the doshas back into a harmonious state through lifestyle interventions, spiritual nurturing, and treatment with herbo-mineral formulas based on one's mental and bodily constitution. (Altern Ther Health Med. 2001;7(2):36-42)

Chinta (serious worry) is an older sister of Chita (cremation bed) because Chita burns a person to ashes only after death but Chinta burns a person alive.

—Anonymous

Ayurveda, a combination of the 2 Sanskrit words *ayu* (longevity) and *ved* (knowledge), is the traditional medical system that has been practiced in India for more than 5000 years. It is a holistic and systematic approach to total healthcare that is not fully understood by many medical professionals in the West. Many in

the scientific, educational, political, and media communities perceive Ayurveda simply as a series of home remedies for housewives, thereby contributing to its frequent omission from historical reviews of the world's medical systems.^{1,2} According to a variety of definitions, including that of the NIH's Office of Alternative Medicine (renamed the National Center for Complementary and Alternative Medicine in 1998),³ Ayurveda is a complementary and alternative medicine and is beginning to be studied and recognized under that rubric.

The goal of this article is to provide an overview of Ayurveda so that medical practitioners and the general public may learn to understand and appreciate it to its full extent. To accomplish this objective, this review will focus on the historical foundation of Ayurveda, its evolution and development, ancient Ayurvedic texts, specialties, and the concepts of disease etiology within Ayurveda. To supplement this review, information regarding diagnosis and treatment of disease within Ayurveda can be found elsewhere.⁴

HISTORICAL PERSPECTIVE: ANCIENT TEXTS

Based on citations in *Rigved* (an ancient Sanskrit text), the traditional system of healthcare in India originated around 4500 to 1600 BC.^{5,6} In the early days, healthcare was considered a part of the spiritual tradition of a universal religion or *dharm*. The concept of disease management using various herbs and dietary interventions evolved into a science that formed the basis of the name "Ayurveda" around 2500 to 600 BC. The early sages conceived the vast knowledge of Ayurveda without animal testing or randomized controlled trials. It is believed that the sages were deeply devoted to God and considered health an integral part of spiritual life, receiving the knowledge of Ayurveda through divine revelation. This knowledge was then transmitted orally to subsequent generations and eventually transcribed into books.

The original 4 books of spiritual knowledge are called *Veds* and include *Rigved*, *Samaved*, *Yajurved*, and *Atharved*. Ayurveda was discussed in all of these books and organized into its own system of healthcare as a branch of the *Veds* called *Upved* (limb of the *Ved*). There are 10572 hymns in *Rigved*, which discusses the 3 *doshas* (biomaterials)—namely, *vata*, *pitta*, and *kapha*—the imbalance of which is believed to be the basis of disease. Organ transplant, plastic surgery, artificial limbs, the use of herbs and minerals to heal diseases of the mind and body, and processes to prolong life also are discussed in the hymns. There are 5977

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hymns in *Atharved* that discuss anatomy, physiology, and surgery.⁷

By 600 BC, 2 schools of Ayurveda had formed: the School of Physicians (the *Atreya* school) and the School of Surgeons (the *Dhanvantari* school), similar to today's allopathic organization of medicine. These schools did a considerable amount of work that established a strong empirical foundation for the practice of Ayurveda and dispelled the mystery and superstition surrounding its origins by divine revelation. Consequently, Ayurveda grew into a respected and widely used system of healing in India. People from many countries including China, Tibet, Greece, Italy, Egypt, Afghanistan, and Iran came to Indian Ayurvedic schools such as Nalanda University and Kashi University to receive formal training in Ayurveda and to propagate it worldwide.⁸

EARLY AYURVEDIC LITERATURE

The 3 most important treatises in Ayurveda are *Charak Samhita*, *Susrut Samhita*, and *Ashtang Hridaya Samhita*, also known as *Vridhdha Triya* (the Senior Triad). Next in line are *Madhav Nidan Samhita*, *Sarangdhar Samhita*, and *Bhavprakash Samhita*, known as *Laghu Triya* (Junior Triad).

Charak Samhita

Charak (100-400 AD),⁹ the leader of the *Atreya* School of Physicians, first compiled the knowledge of Ayurveda known as *Charak Samhita*. In *Charak Samhita*,⁸ Ayurveda is described as follows:

That is named the science of life, wherein are laid down the good and the bad of life, the happy and the unhappy life, and what is wholesome and what is not wholesome in relation to life, as also the measure of life. The mind, spirit, and body are together, as it were the tripod; the world endures by reason of cohesion and on that are all things established. The aggregate of the mind, spirit and body is man; he is the conscious agent. He is regarded as the subject matter of this science, and it is indeed for his sake that this science has been promulgated. The action here is establishing the equilibrium of the body-elements and the procedure of maintaining the equilibrium of the body elements is the objective of this science. The body and that which is called the mind are both considered to be abodes of disease, likewise of well-being is harmony.

From this description, it is clear that the mind and body are integral parts of the health and disease equation in Ayurveda; neither can be treated successfully in isolation from the other. *Charak Samhita* states that the first cause of illness is loss of faith in the Divine. The external causes of illness include diet, lifestyle, and exposure to chemical, physical, and biological agents. A large section of the *Charak Samhita*, a sacred text, deals with activities of daily living, diet, pathogenesis and treatment of various diseases, reversal of the aging process, and aphrodisiacs. The disease classification in *Charak Samhita* includes more than 200 diseases and 150 pathological conditions and congenital defects, and is based primarily on body organ systems and physiological func-

tions. The classification fits well into the disease categories based on World Health Organization criteria. Two categories of diseases and conditions are given below as examples of this remarkable text, authored more than 1200 years ago.⁹

1. Infectious and parasitic diseases including *Sannipat jwar* (typhoid fever), *Vismajwar* (recurrent fever), *Vishuchika* (cholera), *Visarpa* (acute spreading suppuration), *Kami* (parasitic infections), *Slipad* (elephantiasis), *Masurika* (small pox), *Kaksa* (herpes), *Dadru* (ringworm), *Vicharchika* (scabies), and *Dhanustambha* (tetanus)

2. Neoplasia such as *Arbud* (nonsuppurating swelling), *Galganda* (tumor on the side of the neck), *Vradhna* (permanent swelling), *Kakana* (malignant growth with diagnostic criteria), *Granthi* (glandular swelling), *Vidardika* (swelling in the groin), *Gulma* (hard swelling), *Rakta Granthi* (tumor in the bladder neck), and *Vata Sthila* (hard tumor)

Following are the other categories of disease listed in the *Charak Samhita*, with the number of conditions in each category in parentheses:

3. Endocrine, metabolic, nutritional, and immunological disorders (8)
4. Diseases of blood and blood-forming organs (4)
5. Mental disorders (4)
6. Diseases of the nervous system and senses (33) and eye (3), ear (5), and nose (9)
7. Diseases of the circulatory system (6)
8. Diseases of the respiratory system (7)
9. Diseases of the digestive system (31)
10. Diseases of the genitourinary system (12)
11. Complications of pregnancy (4)
12. Diseases of skin and subcutaneous tissue (23)
13. Diseases of the musculoskeletal system and connective tissue (16)
14. Congenital anomalies (2)
15. Certain conditions originating in the perinatal period (*Nadirog*)
16. Symptoms and signs of ill-defined conditions

Susrut Samhita

The *Dhanvantari* school of surgeons, represented by *Susrut Samhita* (300-400 BC),¹⁰ offers detailed and sophisticated discussions of surgical equipment, classification of abscesses, burns, fractures, wounds, and amputation, as well as plastic surgery (ie, rhinoplasty) and anal/rectal surgery similar to that of modern medicine.¹⁰ *Susrut Samhita* gives a complete description of human anatomy, including bones, nerves, blood vessels, and the circulatory system, and mentions the brain as the center of the senses. It also describes anatomical dissection and surgery that was practically nonexistent in contemporary Western culture at that time. Surgical procedures on the skull are also mentioned, including a *tijak* (lateral) incision on the temple to alleviate headache and a procedure for the removal of maggots from infected wounds. The description of month-to-month development of the fetus given in *Susrut Samhita* is remarkably similar to that known today. The

embryology of a 4-month-old fetus is described, including development of the heart, abdomen, and liver.

Susrut Samhita also includes a comprehensive chapter on therapeutics (*Uttar Tantra*) that deals with general diseases (eg, diarrhea, lung diseases, and fever) and describes the science of massage, acupressure, and acupuncture using *marma* (sensitive or vital points). This demonstrates that extensive knowledge of medical and surgical procedures existed in Ayurveda nearly 2000 years ago.

Susrut Samhita also delineates Ayurveda into 8 distinct branches:

1. *Kaya Chikitsa*, internal medicine
2. *Bal Chikitsa*, pediatrics
3. *Graha Chikitsa*, psychiatry
4. *Shalya Tantra*, surgery
5. *Shalakyata Tantra*, ophthalmology, otorhinology, and laryngology
6. *Agad Tantra*, toxicology
7. *Rasayan Tantra*, geriatrics (longevity and rejuvenation)
8. *Vajikarana Tantra*, eugenics and aphrodisiacs

Ashtang Hridaya Samhita

The third major treatise in Ayurveda, written by Vagbhata in 700 AD, is called *Ashtang Hridaya Samhita*.⁶ It is a concise version of the works of *Charak* and *Susrut*, which are discussed above. These 3 important books in Ayurveda are known collectively as the *Vridhha Traya* (Senior Triad). Ayurveda is considered the oldest medical system in the world, because it is extensively described in *Rigved*, dated between 4500 BC to 1600 BC, and the complete medical texts are still extant. No other system of medicine, including traditional Chinese medicine, has such a complete and well organized set of recorded medical texts that are as old as the *Rigved*.

The *Laghu Triya* (Junior Triad) of Ayurveda includes (1) the *Madhava-nidana samhita* (diagnosis of disease) by Madhavkar, written in 1200 AD; (2) *Sarangdhar Samhita* (materia medica), written by Sarangdhar in 1400 AD; and (3) *Bhavprakash Samhita* (materia medica), by Acharya Bhav Mishra, written in 1500 AD. The other equally important books are *Kashyap Samhita*, which gives a comprehensive description of *Kaumara-bhritiya Tantra* (maternity and child care); *Chakradatta* and *Dravyaguna-samagraha*, compiled in 1100 AD by Chakrapani, the renowned physician who wrote an excellent commentary on the Senior Triads and conducted pioneering works on combination therapy with various herbs and minerals; and *Raj-nighantu* by Narhari Pandit and *Madanpala-nighantu* by Madanpala, the most notable books on medicinal plants.

In Ayurvedic texts, 50 distinct pharmacological categories of medicinal plants are described. The categories include anti-inflammatory, analgesic, antiallergic, antihistaminic, antidiuretic, diuretic, antiemetic, emetic, purgative, astringent, antiasthmatic, antipyretic, and antidotal. The full list of categories is shown in the Table to demonstrate the extensive knowledge and understanding of pharmacology in Ayurveda at that time.⁶ All the Ayurvedic texts cited above are available in print and are being taught in Ayurvedic medical schools in India.

Classification of Ayurvedic herbs in *Charak Samhita**

Ayurvedic name	English name
1. <i>Jivaniya</i>	Vitality promoter
2. <i>Brihaniya</i>	Body mass promoter
3. <i>Lekhaniya</i>	Emaciating
4. <i>Bhedaniya</i>	Excretion promoter
5. <i>Sandhaniya</i>	Bone fracture healer
6. <i>Dipaniya</i>	Appetite promoter
7. <i>Balya</i>	General tonic
8. <i>Varnya</i>	Complexion refiner
9. <i>Kanthya</i>	Voice refiner
10. <i>Hridya</i>	Heart tonic
11. <i>Triptighna</i>	Antisaturative
12. <i>Arshoghna</i>	Antihemorrhoid
13. <i>Kushthaghna</i>	Antidermatosis
14. <i>Kandughna</i>	Anti-itch
15. <i>Vishaghna</i>	Antidote
16. <i>Stanyajanana</i>	Breast milk producer
17. <i>Stanyashodhana</i>	Breast milk purifier
18. <i>Sukrajanana</i>	Sperm cell producer
19. <i>Sukrasodham</i>	Semen purifier
20. <i>Snehopaga</i>	Oily lubricant
21. <i>Swedopaga</i>	Diaphoretic
22. <i>Vamanopaga</i>	Emetic
23. <i>Virechanopaga</i>	Purgative
24. <i>Asthapanopaga</i>	Enema
25. <i>Anuvasanopaga</i>	Medicated oil enema
26. <i>Sirovirechanopaga</i>	Errhine
27. <i>Chhardinigravana</i>	Antiemetic
28. <i>Hikkanigraha</i>	Antihiccup
29. <i>Trishananigraha</i>	Antidyspeptic
30. <i>Svayathuhara</i>	Anti-inflammatory
31. <i>Jwarahara</i>	Antipyretic
32. <i>Shramahara</i>	Antifatigue
33. <i>Dahaprashamana</i>	Body coolant
34. <i>Sheetaprashamana</i>	Antibody coolant
35. <i>Udardaprashamana</i>	Antiallergic
36. <i>Angamardaprashamana</i>	Body pain reliever
37. <i>Shulaprasamana</i>	Antispasmodic
38. <i>Shonitasthapana</i>	Blood tonic
39. <i>Vedanasthapana</i>	Analgesic
40. <i>Purishasangrahaniya</i>	Astringent
41. <i>Sangyasthapana</i>	Resuscitative
42. <i>Prajasthapna</i>	Fertility promoter
43. <i>Vayahsthapna</i>	Age-sustainer
44. <i>Swasahara</i>	Antiasthmatic
45. <i>Kasahara</i>	Anticough
46. <i>Mutravirechaniya</i>	Diuretic
47. <i>Mutravirajaniya</i>	Urine pigment reducer
48. <i>Mutrasangrahaniya</i>	Antidiuretic
49. <i>Pusishavirajaniya</i>	Fecal pigment reducer
50. <i>Krimighana</i>	Anthelmintic

* Adapted from Chatterjee and Pakrashi⁶

The practice of Ayurveda declined considerably during the foreign invasion of India by Mohammedans from 1500 to 1800 AD.⁶ Ayurvedic books were destroyed and its practice was largely abandoned in favor of the invaders' own system of medicine. Several Europeans—Sirs Jones, Flemmings, and Ainslie (1813); and Roxburgh (1834)—began compiling available literature on Ayurvedic herbs around 1800. The first *Pharmacopoeia India* was completed by Warren in 1868, and later supplemented by Mohideen Sheriff (1869) and Dymock, Warden, and Hooper (1890-1893). The landmark compendium, *The Dictionary of Economic Products of India*, prepared by George Watts between 1889 and 1904, greatly facilitated the use of Indian medicinal plants. This text discusses the earlier works (1813-1893) concerning medicinal plants and includes plant habitat, use of appropriate parts of the plant, cultivation, economic importance, yield of the drug, and results of clinical trials when available.

The credibility of Ayurveda declined prior to the 19th century due to unsubstantiated claims made by Ayurvedic practitioners and adulterations or substitution of herbs by unqualified physicians. A significant obstacle for the scientific credibility of Ayurveda was the lack of standardization (ie, the identification of plants, collection of the proper parts of the plant, and exact ingredients in a complex formula). Climatic and seasonal variations may produce changes in chemical constituents of plants. These changes and other ecological factors further muddle the issue of standardization. Other problems that affected Ayurvedic medicine include the adherence to traditional methods for preparing formulas that make toxic metals such as mercury nontoxic and the determination of the stability of preparations.

INFLUENCE OF AYURVEDA ON EASTERN AND WESTERN COUNTRIES

The translation of the Ayurvedic texts into Chinese occurred around 400 AD. By 700 AD, many Chinese scholars (eg, Hiuen [629 AD] and Itsing [622-688]) came to Nalanda University in India to study Ayurvedic medicine.⁵ Knowledge of Ayurveda, coupled with the spirituality and philosophy of Buddhism, greatly influenced Chinese medicine. In 800 AD, Ayurvedic books were translated into Arabic. In 900 AD, Islamic physicians such as Avicenna and Razi Serapion quoted Ayurvedic texts, became very influential, and contributed to the evolution of medicine in Europe. In 1600 AD, Paracelsus, known as the father of modern Western medicine, practiced and propagated a system of medicine that borrowed heavily from Ayurveda,¹¹ indirectly, through Arabic medicinal tradition as well as Greek and Near Eastern sources.⁵

DISEASE ETIOLOGY (TRIDOSHAS)

Men and women are made up of 2 components: mind and body. The mind is made up of 2 psychic components—*atma* (soul, spirit) and *nana* (mood, cognition)—whereas the body is made of 3 somatic components: *dhatu*s (body tissues), *malas* (waste products), and *doshas* (body materials, or psychic abnormality).^{12,13} Health is the balanced state and perfect functioning of

all mind and body components. A disease is an unbalanced state and/or imperfect functioning of mind and body components. There are 2 kinds of *doshas*: *saririk* (somatic), which include *vata*, *pitta*, and *kapha* (biomaterials); and *manasik* (psychological), which include *satogun*, *rajogun*, and *tamogun*. A close interdependence exists between the *manasik* and *saririk doshas*: if one is disturbed, the other is also disturbed. The abnormal disturbance of *vata*, *pitta*, and *kapha* (known collectively as the *tridoshas*) is considered the major factor in the causation of disease.

Dhatu (Body Tissues)

*Dhatu*s are the supportive tissues of the body. There are 7 *dhatu*s described in *Charak Samhita*: (1) *rasa*, plasma; (2) *rakta*, blood; (3) *mansa*, muscular tissue; (4) *asthi*, bone and osseous tissue; (5) *meda*, adipose tissue; (6) *majja*, marrow and myeloid tissue; and (7) *sukra*, reproductive tissue. Secondary tissues are known as *updhatus*. *Dhatu*s and *updhatus* make up the physical bulk of the body. All *dhatu*s are made up of minute *parmanu* (cells) of different structure and function, forming the different organs of the body. Each cell is connected with another cell with *suksma srotas* (minute channels or pores), through which cells receive and distribute nutrients and metabolic waste products. This description of cellular communication, written more than 1200 years ago, is consistent with modern understanding of intercellular exchange. *Dhatu*s contain a material called *ojas* (activator, essence, or hormones), which possesses specific properties and functions (for example, a radiating personality).

The word *mala* means waste products of the body, and includes 2 kinds: *ahar mala*, or food waste (feces and urine) and *dhatu mala* (cellular metabolic waste, sweat). These waste products promote the maintenance of health as long as they are within normal limits.

Saririk Doshas (Somatic Materials)

According to *Charak Samhita*, *vata* suggests speed, movement, adjustment, and conduction of particles and odor. The physical properties of *vata* are described as dry, cold, light, penetrating, mobile, transparent, and rough. *Vata*, in conjunction with *shirish* (the brain), *sushumna* (the spinal chord), *panchvayu* (the voluntary and involuntary nervous systems), and *nadi* (the motor and sensory systems), performs the functions of the central, autonomic, and peripheral nervous systems. As such, it controls the respiratory, blood, lymphatic, excretory, and reproductive systems, as well as all types of movements. It is also responsible for the cognitive and neocognitive behavior of the brain and internal secretory glands and includes various chemical neurotransmitters and hormones. *Charak Samhita* further describes the following 6 main chakras (extrasensory channels) of *vata*, which correspond to modern autonomic nervous system plexuses:

1. *Agnya chakra*, nasociliary extension of the cavernous plexus of the sympathetic nervous system
2. *Vishuddha chakra*, pharyngeal plexus of the sympathetic nervous system
3. *Anahat chakra*, cardiac plexus of the sympathetic nervous system

4. *Manipur chakra*, plexus of the cholinergic nerves
5. *Swadhistan chakra*, hypogastric plexus
6. *Muladhar chakra*, pelvic plexus

Vata cooperates and controls *pitta* and *kapha* to accomplish various physiological functions. There are 5 *vatas* in the body according to their location and function: (1) *pran*, (2) *apan*, (3) *vyan*, (4) *udan*, and (5) *saman*. *Pran vata* is situated in the central nervous system and some silent areas of the brain that are not yet identified anatomically but have been treated and discussed at length by yoga philosophy scholars. *Pran vata* coordinates the action of other *vatas*. *Apan vata* is chiefly located in the lumbosacral region, but also distributed over the gluteal, genital, urinary, inguinal, and visceral regions. It regulates discharge of seminal fluid, menstrual fluid, feces, and urine, and aids in childbirth. *Vyan vata* is located in *hradaya* (the fluid pumping system) and controls contraction and relaxation of muscles. It consists of all the unconscious reflexes produced along the spinal cord (corresponding to the autonomic nervous system) and participates in blood circulation, lymph circulation, sweating, and other secretions of the body through the stimulation of nerves. *Udan vata* is situated in the chest, *nabhi* (umbilical area), and nasal and pharyngeal passages. It controls speech, respiration, energy, body luster, and consciousness, and influences the respective centers in the brain. *Saman vata* is distributed over the entire abdomen and controls digestion and assimilation of food, as well as excretion of waste products.

The symptoms of deranged *vata* are as follows: irregular movement of the limbs, neuralgia, throbbing and tearing pain, suppression of secretions, contraction of muscles (spasms), roughness of the skin, formation of bone cavities, suspension of bowel movements, and astringent taste in the mouth. *Vata* disturbance may be caused by overworking, grief, worry, lack of sleep, intentional retention of bodily waste, excess exercise, or sexual indulgence, as well as ingestion of bitter, pungent, astringent, or dry foods.

The word *pitta* indicates *agni* (fire). It is described as hot, sharp, sour, pungent, and moist. It is responsible for appetite, thirst, digestion, assimilation of nutrients, production of body heat, normal eyesight, softness of the body, luster, mental calmness, and intelligence. Ingestion of salty, hot, irritating, or sharp-tasting foods, as well as anger, may cause disturbance in *pitta*. Symptoms of deranged *pitta* are a burning sensation, restlessness, an overactive or less active digestive process, sweating, intoxication, and a pungent or sour taste in the mouth. There are 5 types of *pittas* based on their location and function: (1) *pachak*, (2) *ranjaka*, (3) *sadhak*, (4) *alochak*, and (5) *bhrajak*.

Pachak pitta is located between the stomach and small intestine and corresponds to bile and pancreatic juices. *Ranjaka pitta* is located in the stomach, duodenum, liver, spleen, and pancreas, and is responsible for digestion and metabolism to provide nutrients, blood constituents, and the formation of the blood cells in the spleen. *Sadhak pitta* is located in the heart as *prana vata* and is responsible for life span, complexion, energy, body temperature, and body luster. *Alochak pitta* is located in the retina and is responsible for vision. *Bhrajak pitta* is situated in the skin and is responsible for the heat, moisture, and health of the skin, as well as the

vasomotor mechanism and adaptation to the surrounding environment. The symptoms of deranged *pitta* are burning sensation, redness, digestive system disturbances, excessive sweating, fainting, symptoms of intoxication, pungent and sour taste, and the inability to see white and dark red color.

The word *kapha* in *Charak Samhita* suggests attachment and assimilation. It is described as a moist, steady, cool, heavy, soft, and slimy material. It is responsible for normal moisture, stability of the joints, firmness of the body, a proportionate bulk and weight, manhood, strength, endurance, and courage. *Kapha* is opposite to *pitta* in function and thus balances the overall functions of *pitta*. For example, *pitta* generates heat whereas *kapha* provides ways to eliminate the heat through the skin, lungs, urine, and feces. *Kapha* is composed of the lymphatic system, body fat, muscle mass, and mucous-producing systems. There are 5 types of *kaphas* based on their location and function: (1) *avalambak*, (2) *kledak*, (3) *bodhak*, (4) *tarpak*, and (5) *shleshaka*.

Avalambak kapha is situated in the chest and enables other *doshas* to perform their function by supplying humidity or body-water, and consists of the liquid portions of the blood, intracellular fluid, and interstitial fluid. *Kledak kapha* is situated in the stomach and is described as liquid, mucoid, sweet, and cooling in nature; it moistens the bolus. *Bodhak kapha* is located under the tongue and consists of saliva and other juices that moisten the mouth; it helps the bolus descend and creates taste, working hand in hand with *kledak kapha*. *Bodhak* and *kledak kaphas* work together by mechanically propelling the bolus and chemically by enzymatic conversion of starch into sugar. *Tarpak kapha* is situated in the brain and sustains the 5 special senses. *Shleshaka kapha* is located in the joints. It corresponds to the synovial areas of modern anatomy, holds the joints together, and provides lubrication.

The symptoms of deranged *kapha* are excessive moisture, hardness of tissues, itching, feeling cold and heavy, low secretory activity, mucous membranes heavily coated with mucus, reduced activity of limbs, swelling, weak digestive power, sleepiness, pale skin, salty taste, and slow recovery after an illness.

Manasik Doshas (Psychological Attributes)

Charak Samhita describes common human attributes such as desire for pleasure, aversion to pain or misery, intelligence, mind, will, logical faculty (*vicarna*), memory, knowledge gained from experience (*vijnana*), and ability, all of which differ in characteristics according to the 3 *manasik doshas*: *satogun*, *rajogun*, and *tamogun*. Manifestations of *satogun* (godly) are compassion, self-control, forbearance, devotion to the truth, proper conduct, faith in God, knowledge gained from self-realization (*jnana*), intelligence, resolution, comprehension, and nonattachment. Manifestations of *rajogun* (kingly) are sensitivity to pain, restlessness of mind, lack of true understanding, egoism, lack of honesty, lack of compassion, false pride, overconfidence, buoyancy of spirit, strong desires, and anger. Manifestations of *tamogun* (evil) are despondency, agnosticism, and atheism, tendency to evil conduct (*adharma*), dull intellect, lack of knowledge and insight, perverted mind, physical and mental inertia, and somnolence. *Manasik*

prakritis (mental constitutions) are characterized according to the preponderance of the manifestations as *sattvik* (godly), *rajasik* (kingly), or *tamsik* (evil), as discussed under constitutions.

Prakritis (Constitutions)

Prakritis (constitutions) is the physical or psychological profile of a person according to the predominance of a *manasik* or *saririk dosha*. *Manasik prakriti* (mental or psychological constitution) is essentially the inherent personality or psychological attributes of a person determined by the predominance of one of the *manasik doshas*. *Charak Samhita* classifies 3 types of personalities: (1) *sattvik* (godly), (2) *rajasik* (kingly), and (3) *tamasik* (evil). Briefly, *sattvik* constitution indicates a pure minded, compassionate, loving, and righteous person capable of enduring all degrees of hardship and pain through willpower. *Rajasik* indicates an egoistic, ambitious, proud, competitive, controlling, and power-hungry person who may be made to submit to pain and unpleasant medical treatment through persuasion and logic. *Tamasik* indicates a lazy, sleepy, depressed, greedy, irritable, irresponsible, and selfish person with a large appetite who is horrified at the prospect of bodily pain. It is important for Ayurvedic practitioners to consider these characteristics when pondering treatment options.

Saririk prakriti (body constitution) is determined by the inherent natural predominance of one or more of the *doshas* as described above. *Charak Samhita* describes 4 major types of constitutions: (1) *vata prakriti*, (2) *pitta prakriti*, (3) *kapha prakriti*, and (4) *sama prakriti* (a person with balanced *doshas*). Subcategories of *prakritis* also exist, depending on the degree of predominance of each of the 3 *doshas*. However, for all practical purposes the first 3 categories are considered sufficient for the diagnosis and treatment of illnesses and are described here.

A person with *vata prakriti* demonstrates a low body weight, light bone structure, and a predominant Adam's apple. The person is enthusiastic, fast to act, forgetful, artistic, loves esoteric material, and is shy and sensitive. This person is more susceptible to *vata* imbalance diseases such as rheumatism, nervous disorders, sciatica, insomnia, dry skin, and constipation. Other health conditions likely to occur are receding gums, weak bones, infertility, impotence, colic, flatulence, stuttering, ringing in the ears, irregular menstruation with cramps, varicose veins, paralysis, blood clotting, anorexia, shivering fits, and poor blood circulation.¹⁴

A *kapha prakriti* person has a heavy body weight, solid body structure, strength, endurance, and secure feeling, and is self-confident, not easily rattled, and somewhat lethargic. The person prefers others to work for him, has entrepreneurial sense, patience, politeness, generosity, materialistic tendency, greed, passivity, and a tendency to sleep too much and too often. Examples of diseases associated with *kapha prakriti* are nausea, colds, bronchitis, asthma, kidney stones, swollen lymph nodes, benign tumors, dropsy, goiter, lung and breast cancer, fungal infections, digestive system problems, and obesity.¹⁴

A *pitta prakriti* person falls in between the other 2 constitutions, *vata prakriti* and *kapha prakriti*. This person has a healthy body with well-developed muscles, well-functioning metabolism,

an urge for physical activities, and an interest in science and technical subjects. The person has good speaking ability, aggressiveness, hostile tendency, impatience, tendency to baldness, good mental ability, excellent memory, and loves intellectual activities. Examples of diseases associated with *pitta prakriti* are stress-related diseases; high blood pressure; coronary diseases; thrombosis; ulcers; cancer of the stomach, intestines, and skin; psoriasis; inflammation of the lymph system; infectious diseases; inflammation of the spleen; hepatitis; urinary tract infection; and heartburn.¹⁴

CONCLUSION

Ayurveda has slowly and gradually evolved since it was established more than 5000 years ago. The basis for its foundation may be found in several ancient texts, most notably the Senior Triad and the Junior Triad. Over time it has developed into a highly sophisticated and specialized system of medicine that includes surgery, internal medicine, psychiatry, pediatrics, toxicology, geriatrics (longevity and rejuvenation), eugenics, and aphrodisiacs, analogous to modern allopathic medicine. Its knowledge of anatomy, physiology, histology, and pathology is consistent and remarkably congruent with current Western medicine, especially when one considers its age. The concept of constitution (*prakriti*) is important in Ayurveda because treatment is tailored according to each patient's constitution as determined by *tridosha* analysis (*vata*, *pitta*, and *kapha*). This is unique to Ayurvedic medicine, though the notion that physical, mental, and spiritual harmony is necessary to maintain health is becoming increasingly popular in holistic medicine. This review indicates that Ayurveda is a comprehensive healthcare system that is consistent with modern science and should be investigated further by scientists in the West.

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